Design and Analysis of Dual Source Vertical Tunnel Field Effect Transistor

25VL681 Analog VLSI and Device Modelling Lab

M. Tech. VLSI Design

Batch: 4

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OBJECTIVE:

- To **design and analyze** a Dual Source Vertical TFET (DSV-TFET) using **Silvaco TCAD**.
- To enhance the **ON-state current** (I_{on}) by employing a **dual-source tunneling region**.
- To achieve **very low OFF-state current** and a **subthreshold slope below 60 mV/dec**, overcoming MOSFET limits.
- To study how **doping, gate material, dielectric, and device geometry** affect performance.
- To evaluate **I-V characteristics**, **Threshold Voltage**, **Subthreshold Slope**, and I_{on}/I_{o}ff **ratio** for low- power logic applications.

SOFTWARE HARDWARE USED: Silvaco TCAD

OVERVIEW OF THE DEVICE:

1. The presented structure represents a **Dual source vertical Tunnel Field-Effect Transistor (TFET) NAND device** designed and visualized using **Silvaco ATLAS**.
2. An optimally designed Dual Source Vertical Tunnel Field Effect Transistors is proposed and investigated using technology computer aided design simulation. The vertical tunnel FET have dispersal of source channel drain in the vertical direction which **will enhance the scalability of the simulated device**
3. The benefit of the TFET is switching mechanism which is done **by quantum tunnelling** method through a barrier instead of thermionic emission over the barrier as that of conventional we have developed two-dimensional model of single drain with dual source n-type vertical tunnel field effect transistor.
4. Further introduction to an ultra-thin channel among the drain and gate region will makes aggressive improvement in the numerical simulations of **minimum threshold voltage (VT) of 0.15 V and average subthreshold slope of 3.47 mV/decade.**
5. The variation effect in the channel thickness, drain doping, and work function has been simulated and examined by 2D silvaco TCAD tool.

INTRODUCTION

1. With the continuous downscaling of semiconductor devices, traditional MOSFETs face severe limitations such as **increased leakage current, degraded subthreshold slope, and short-channel effects**. To overcome these challenges, advanced device architectures like the **Tunnel Field-Effect Transistor (TFET)** have emerged as promising alternatives for future low-power electronics.

2. The **TFET** operates on the principle of **band-to-band tunnelling (BTBT)**, where carriers tunnel from the valence band of the source to the conduction band of the channel, enabling **subthreshold swings lower than the 60 mV/decade limit of MOSFETs**. This allows for significant **power reduction** while maintaining strong switching characteristics.

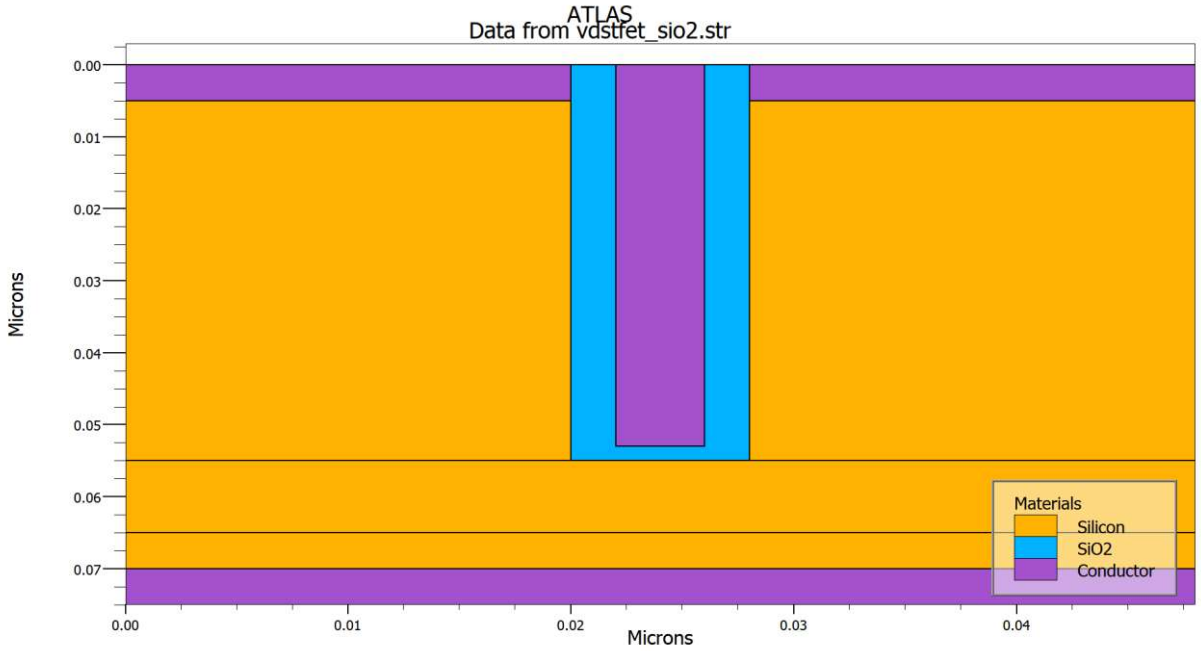
3. In this work, a **dual source vertical Tfet device** is modelled and simulated using **Silvaco ATLAS**. The device consists of three gates — **A normal TFET has one Source and one Drain.**

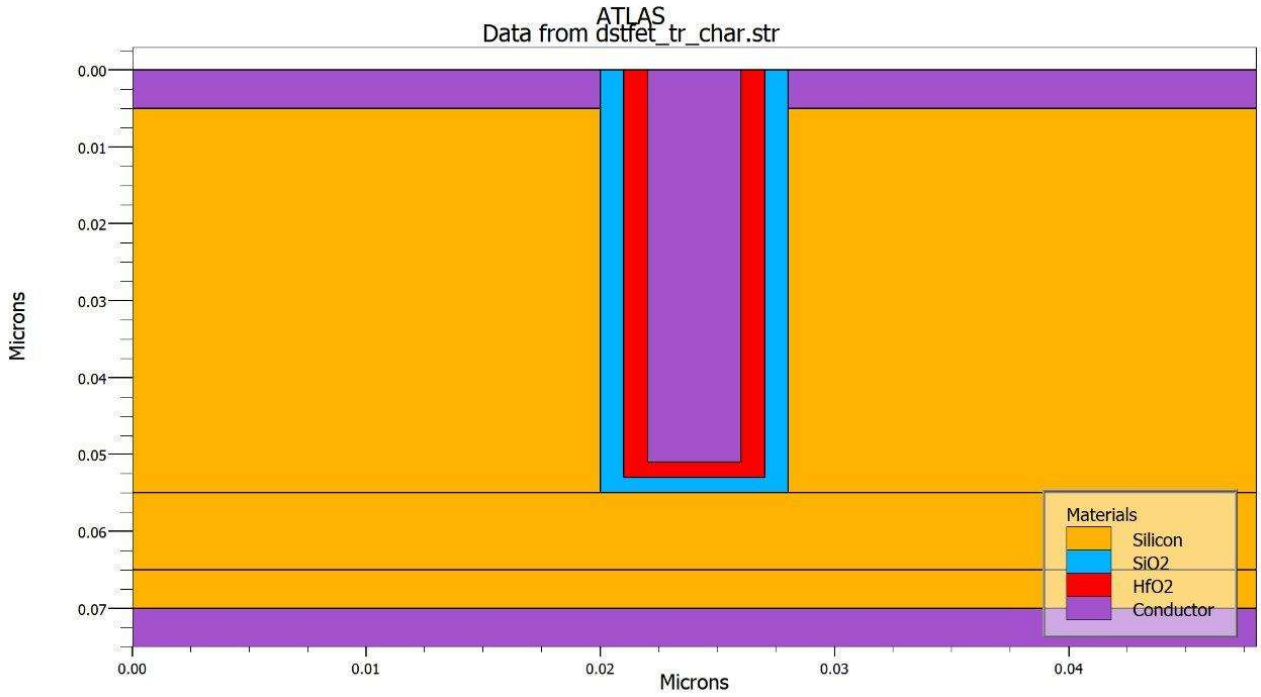
4. **But in Dual Source TFET, the device has two Source regions connected to the same channel. Because adding an extra Source increases the tunneling area.**

More tunneling area → More electron flow → Higher ON current I_{ON}

Hence, **TFETs** are ideal for **ultra-low-power** and **next -generation nano electronic devices.**

STRUCTURE:


ATLAS
Data from vdstfet_sio2.str
Microns
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Materials
Silicon
SiO2
Conductor


ATLAS
Data from dstfet_tr_char.str
Microns
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Materials
Silicon
SiO2
HfO2
Conductor

DEVICE DIMENSIONS AND DOPING CONCENTRATIONS:

Device geometry (cylindrical 3-D nanowire, GAA)

- Gate oxide (SiO_2) thickness (region 2: 0.001 µm): 1nm
- Gate oxide (HfO_2) thickness (region 3: 0.001µm): 1nm

Electrodes

- Gate: legth=40nm
- Source: length =20 nm
- Source : length =20nm
- Drain: length=48nm.

Doping (silicon core, r ≤ 0.010 µm)

- Source (p+): 1×10^{20} cm^{-3}
- Source (p+): 1×10^{20} cm^{-3}
- Channel/body (n): 1×10^{17} cm^{-3}
- Drain (n+): 1×10^{20} cm^{-3}, z = +0.10 → +0.15 µm

*(This is a **p^+–n–n^+** TFET configuration enabling BTBT at the source–channel junction.)*

Biasing used for Id–Vg sweeps

- V_S = 0 V, V_D = 1 V
- Four NAND input cases while sweeping GateCTRL from 0 → 1 V in 0.025 V steps:
  - 00: GateA = 0 V, GateB = 0 V
  - 01: GateA = 0 V, GateB = 0.6 V
  - 10: GateA = 0.6 V, GateB = 0 V
  - 11: GateA = 0.6 V, GateB = 0.6 V

Materials

- Region 1: Silicon (p-type)
- Region 2: Oxide (SiO_2)
- Region 3: Oxide (hfO_2)
- Region 4: Silicon (p-type)
- Region 5: Silicon (n-type)
- Region 6: Silicon (n-type)

Physical models & solver

Model	Purpose	Relevance in DSV-TFET
conmob	Mobility vs doping	Corrects current in highly doped source regions
bgn	Bandgap narrowing	Enhances tunneling at p^+-i junction
srh	Trap-assisted recombination	Controls OFF-state leakage
cvt	Field-dependent mobility	Improves SS accuracy with thin oxide
fermi	Degenerate carrier statistics	Accurate carrier calculation in heavy doping
ni.fermi	Correct intrinsic carrier density	Stabilizes OFF current & band bending
qtunn.dir=ydir	Defines tunneling direction	Required for vertical channel transport
bbt.nonlocal	Main tunneling model	Enables band-to-band tunneling conduction

CODE

```
go atlas
# ---------- VARIABLES  ----------
```

```
# Two oxide layers
set g=0.004
set toxh=0.001
set toxs=0.001
# ---------- MESH ----------
mesh space.mult=1.0
x.m loc=0 spacing=0.005
x.m loc=0.018 spacing=0.005
x.m loc=0.020 spacing=0.0005
x.m loc=0.020+$toxs spacing=0.0005
x.m loc=0.020+$toxs+$toxh spacing=0.005
x.m loc=0.020+$toxs+$toxh+$g   spacing=0.0005
x.m loc=0.020+$toxs+$toxh+$g+$toxh      spacing=0.005
x.m loc=0.020+$toxs+$toxh+$g+$toxh+$toxs spacing=0.005
x.m loc=0.022+$toxs+$toxh+$g+$toxh+$toxs spacing=0.005
x.m loc=0.040+$toxs+$toxh+$g+$toxh+$toxs spacing=0.005
y.m loc=0.0    spacing=0.005
y.m loc=0.005  spacing=0.0005
y.m loc=0.051  spacing=0.001
y.m loc=0.053  spacing=0.001
y.m loc=0.055  spacing=0.001
y.m loc=0.060  spacing=0.005
y.m loc=0.065  spacing=0.0005
y.m loc=0.070  spacing=0.005
y.m loc=0.075  spacing=0.005

# ---------- REGIONS ----------
region num=1 MATERIAL=silicon  x.min=0 y.min=0
region num=2 MATERIAL=sio2    x.min=0.020 x.max=0.020+$toxs+$toxh+$g+$toxh+$toxs
y.max=0.055
```

```
region num=3 MATERIAL=hfo2    x.min=0.020+$toxs x.max=0.020+$toxs+$toxh+$g+$toxh
y.max=0.053

region num=4 MATERIAL=silicon  x.min=0.0   x.max=0.020 y.min=0.005 y.max=0.055
region num=5 MATERIAL=silicon  x.min=0.020+$toxs+$toxh+$g+$toxh+$toxs
x.max=0.040+$toxs+$toxh+$g+$toxh+$toxs y.min=0.005 y.max=0.055
region num=6 MATERIAL=silicon  x.min=0     y.min=0.055 y.max=0.065
region num=7 MATERIAL=silicon  x.min=0     y.min=0.065 y.max=0.070

# ---------- ELECTRODES  ----------
elect num=1 name=gate   x.min=0.020+$toxs+$toxh x.max=0.020+$toxs+$toxh+$g y.max=0.051
elect num=2 name=source x.min=0     x.max=0.020 y.max=0.005
elect num=3 name=source x.min=0.020+$toxs+$toxh+$g+$toxh+$toxs
x.max=0.040+$toxs+$toxh+$g+$toxh+$toxs y.max=0.005
# FIX:  make drain width consistent with silicon span
elect num=4 name=drain  x.min=0     x.max=0.040+$toxs+$toxh+$g+$toxh+$toxs y.min=0.070
y.max=0.075

# extra mesh refinement near corners (optional but harmless)
qtx.mesh loc=0.015 spac=0.0005
qtx.mesh loc=0.050 spac=0.0005
qty.mesh loc=0     spac=0.0005
qty.mesh loc=0.060 spac=0.0005

# ---------- DOPING  ----------
doping uniform p.type conc=1e20 x.min=0 x.max=0.020 y.min=0.005 y.max=0.055
doping uniform p.type conc=1e20 x.min=0.022+$toxs+$toxh+$g+$toxh+$toxs
x.max=0.040+$toxs+$toxh+$g+$toxh+$toxs y.min=0.005 y.max=0.055
```

```
doping uniform n.type conc=1e17 x.min=0 x.max=0.040+$toxs+$toxh+$g+$toxh+$toxs y.min=0.055
y.max=0.065
doping uniform n.type conc=1e19 x.min=0 x.max=0.040+$toxs+$toxh+$g+$toxh+$toxs y.min=0.065
y.max=0.070
# ---------- CONTACTS ----------
contact name=gate workfunction=3.8
contact name=drain
contact name=source
# ---------- MODELS ----------
models conmob bgn srh cvt fermi ni.fermi qtunn.dir=ydir bbt.nonlocal

# ---------- METHOD ----------
method newton trap

# ==============================================================
# DC CHARACTERISTICS  TO GENERATE  LOGS
# ==============================================================

# --- Output 1: Output (Id–Vd) at fixed Vg=1.8 V
solve init
solve vgate=1.8
log outf=dstfet_dr_char.log master
solve vsource=0 vdrain=0 vstep=0.05 vfinal=2.6 name=drain
save outf=dstfet_dr_char.str
log off

# --- Output 2: Transfer (Id–Vg) at Vd=2.6 V
solve init
solve vdrain=2.6
```

```
log outf=dstfet_tr_char.log master
solve vsource=0 vgate=0 vstep=0.05 vfinal=1.8 name=gate
save outf=dstfet_tr_char.str
log off

# --- Output 3: Transfer (Id–Vg) at Vd=0.05 V  (for DIBL Vt_low)
solve init
solve vdrain=0
solve vdrain=0.01
solve vdrain=0.02
solve vdrain=0.03
solve vdrain=0.04
solve vdrain=0.05
log outf=dstfet_20_Vd_005.log master
solve name=gate vgate=0 vstep=0.05 vfinal=1
save outf=dstfet_20_Vd_005.str
log off

# --- Output 4: Transfer (Id–Vg) at Vd=1.0 V  (for DIBL Vt_high)
solve init
solve prev
solve vdrain=0
solve vdrain=0.05
solve vdrain=0.1
solve vdrain=0.2
solve vdrain=0.4
solve vdrain=0.6
solve vdrain=0.8
solve vdrain=1
log outf=dstfet_20_Vd_1.log master
```

```
solve name=gate vgate=0 vstep=0.05 vfinal=1
save outf=dstfet_20_Vd_1.str
log off

# (optional viewers)
#tonyplot dstfet_20_Vd_005.log
#tonyplot dstfet_20_Vd_1.log
#tonyplot dstfet_dr_char.log
tonyplot dstfet_tr_char.log

# ================================================================
# EXTRACTION  (fixed init, no weird spaces)
# ================================================================
go atlas

# Vt at low Vd = 0.05 V
extract init infile="dstfet_20_Vd_005.log"
extract name="vt1" x.val from curve(v."gate", log10(abs(i."drain"))) where y.val=-13.8

# Vt at high Vd = 1.0 V
extract init infile="dstfet_20_Vd_1.log"
extract name="vt2" x.val from curve(v."gate", log10(abs(i."drain"))) where y.val=-5.3

# DIBL
extract name="dibl" ($vt1-$vt2)/(1.0-0.05)

# Subthreshold slope, IOFF, ION,  and a clean Vt from the transfer at Vd=2.6 V
extract init infile="dstfet_tr_char.log"
extract name="subvt" 1.0/slope(maxslope(curve(v."gate", log10(abs(i."drain")))))
extract name="Ioff_n" y.val from curve(v."gate", i."drain") where x.val=0
```

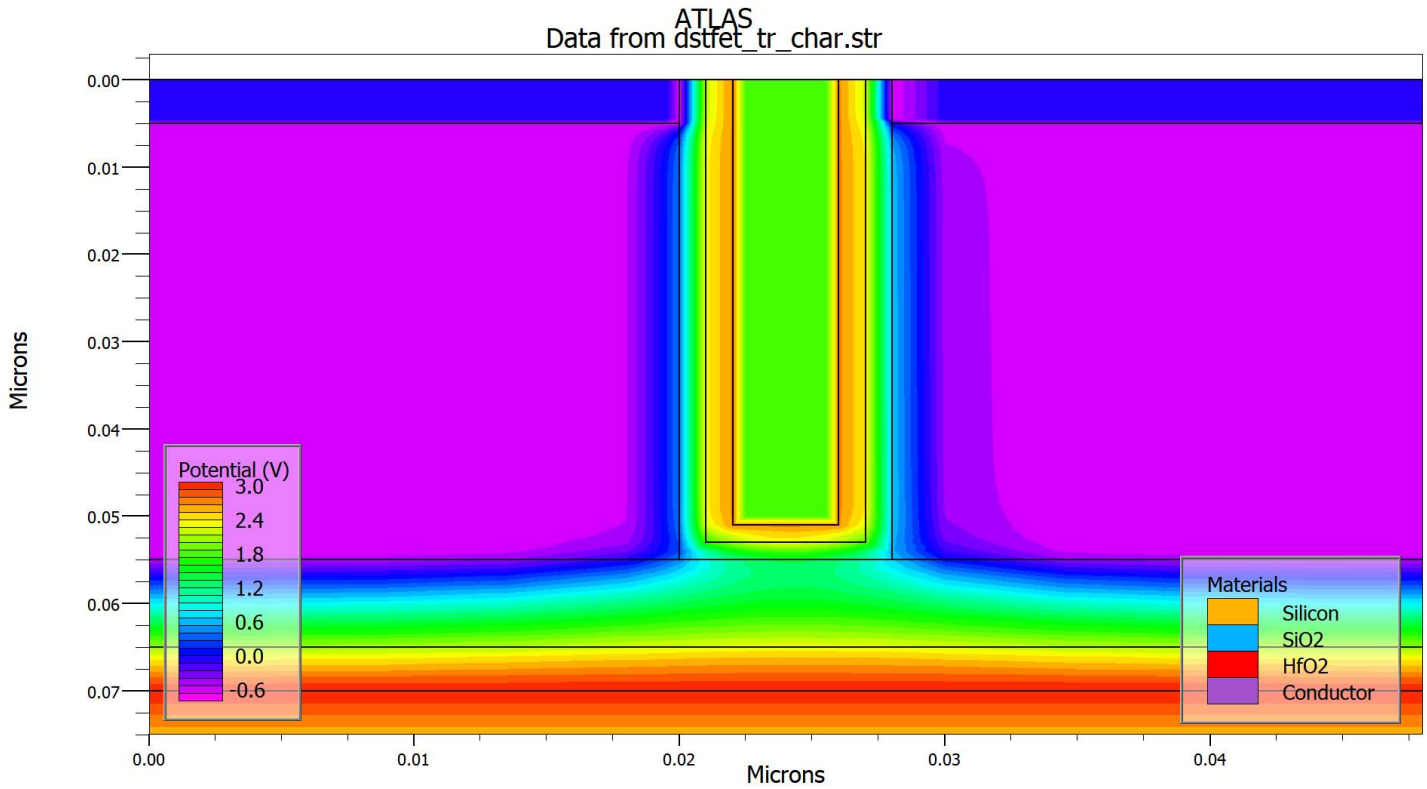
```
extract name="Ion_n"  y.val from curve(v."gate", i."drain") where x.val=1
extract name="Ion_Ioff" $Ion_n/$Ioff_n
extract name="vt" x.val from curve(v."gate", log10(abs(i."drain"))) where y.val=-5.3

Quit
```

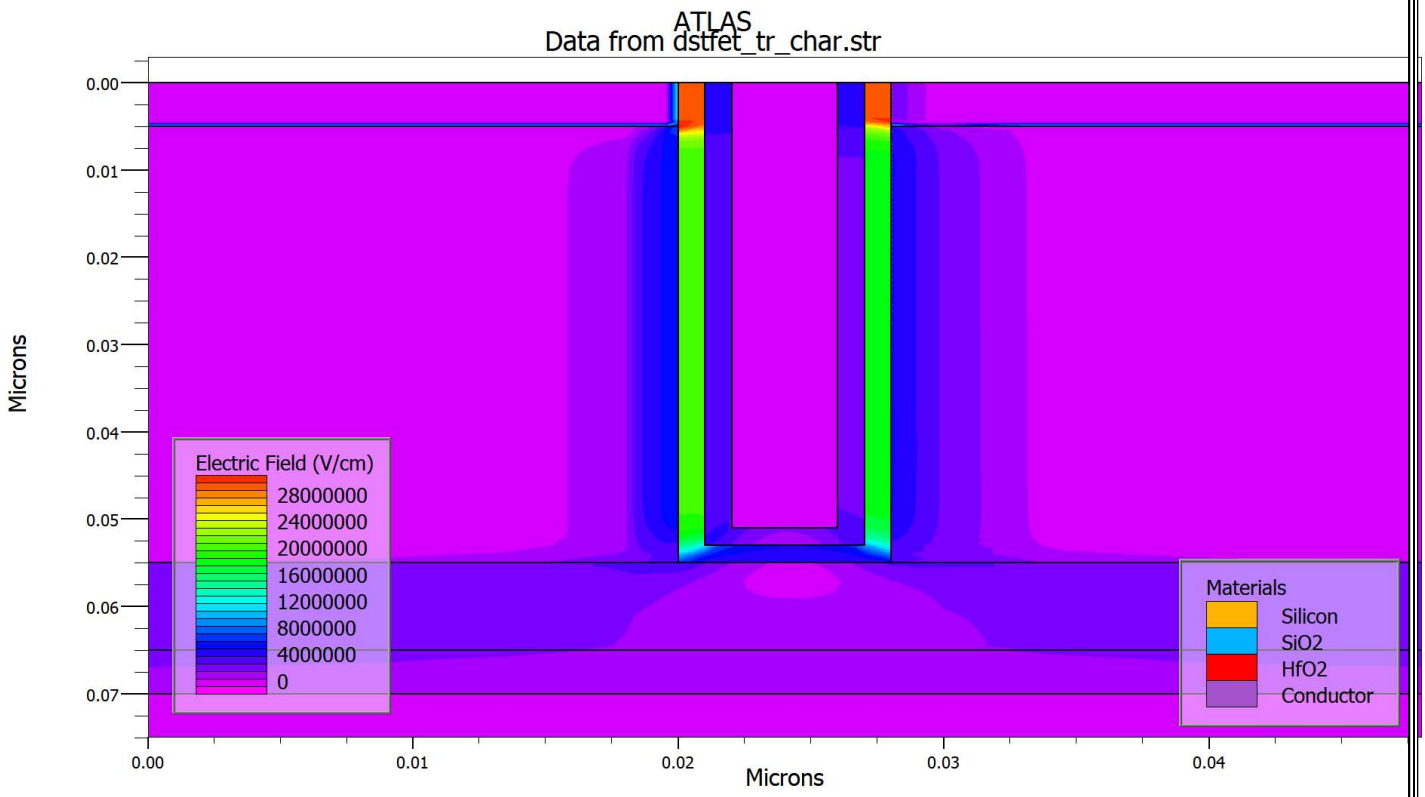
RESULTS:

g	0.004	(# 5)
toxh	0.001	(# 6)
toxs	0.001	(# 7)
vt1	**0.647768578794791**	**(# 135)**
vt2	**0.296141687026775**	**(# 139)**
dibl	**0.370133684210526**	**(# 142)**
subvt	**0.292310658464679**	**(# 146)**
Ioff_n	**4.671190661e-17**	**(# 147)**
Ion_n	**6.228148826e-14**	**(# 148)**
Ion_Ioff	**1333.31121191816**	**(# 149)**
vt	**0.647768578794791**	**(# 150)**

Second Order Effects

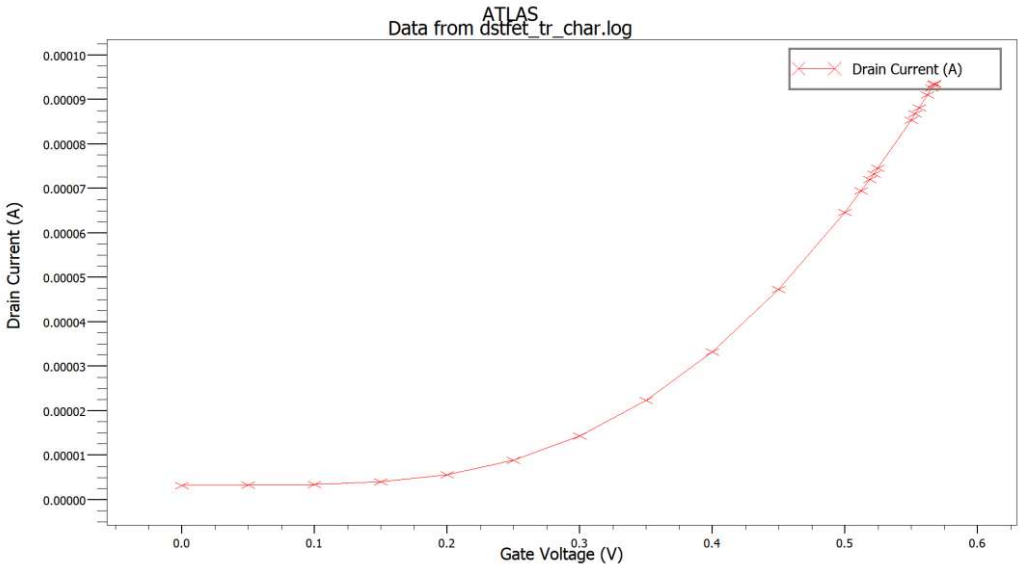

ATLAS
Data from dstfet_tr_char.str
Microns
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Potential (V)
3.0
2.4
1.8
1.2
0.6
0.0
-0.6
Materials
Silicon
SiO2
HfO2
Conductor
0.00
0.01
0.02
0.03
0.04
Microns

Potential

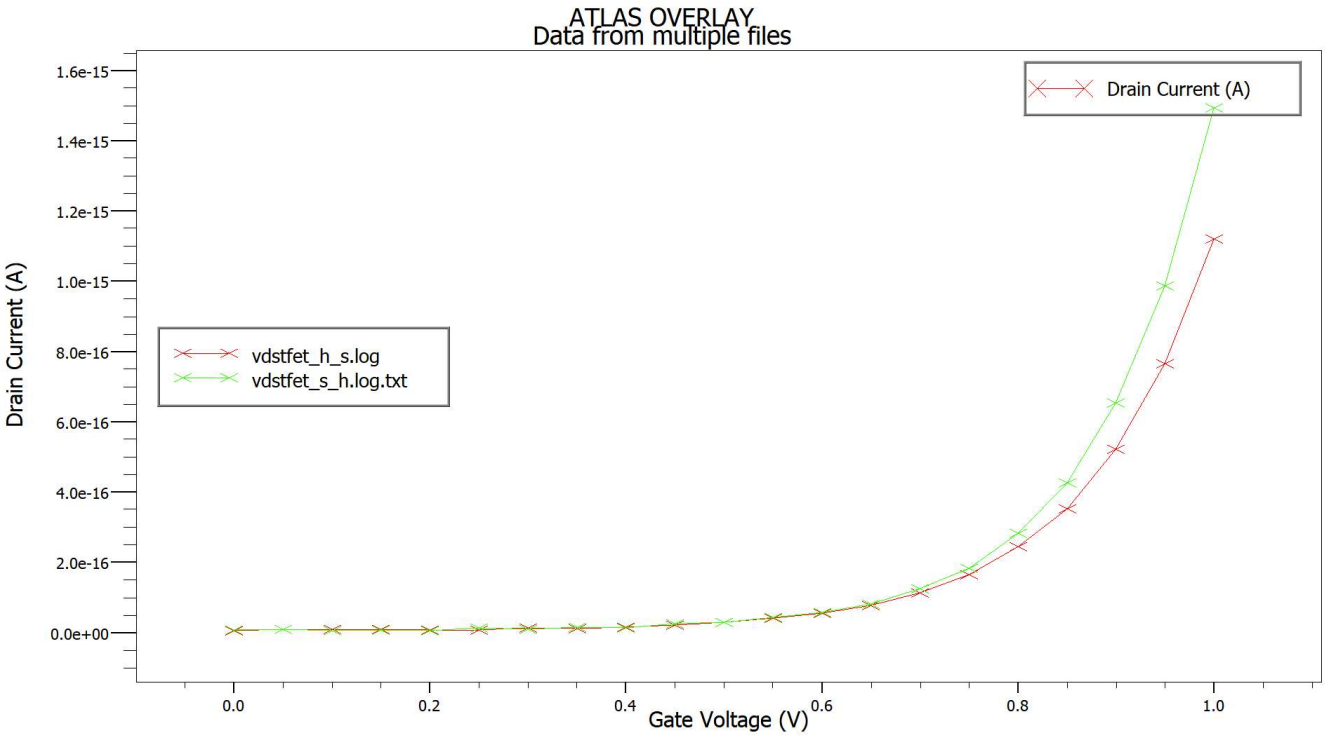


Electric Field

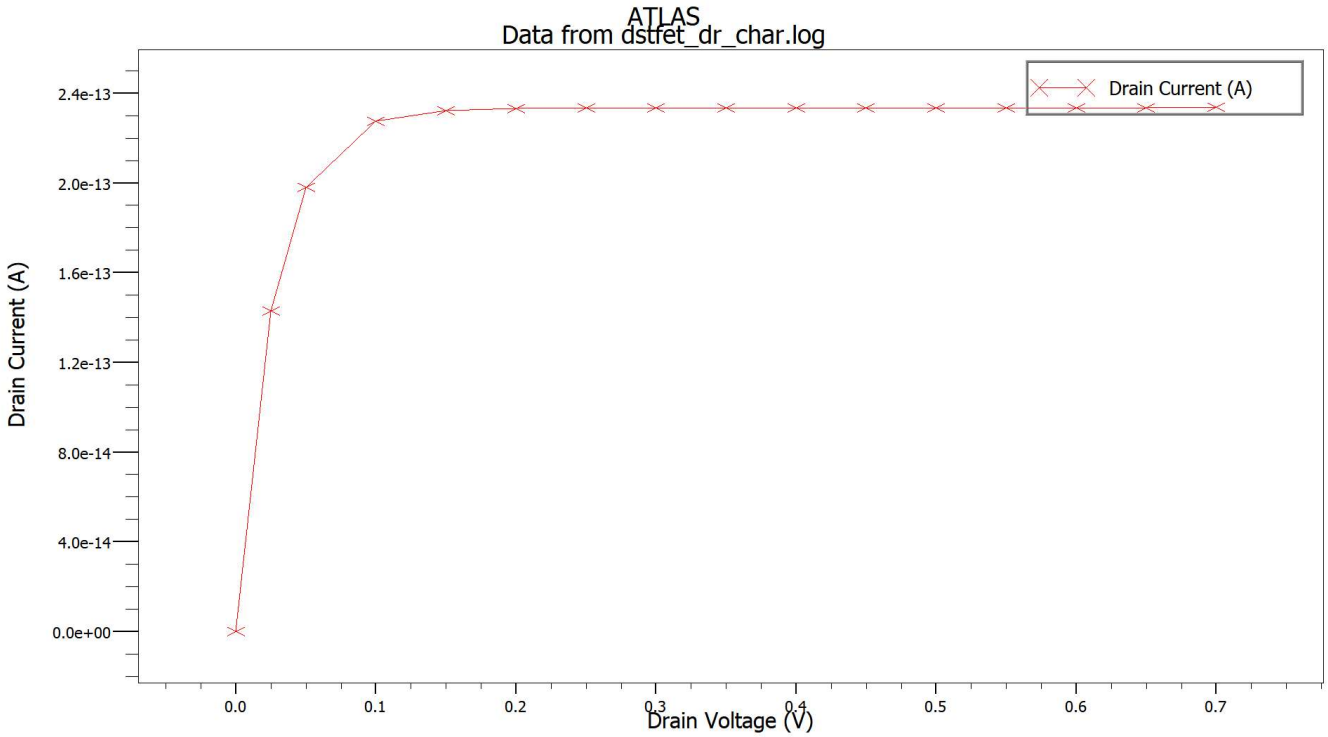
I-V CURVE:



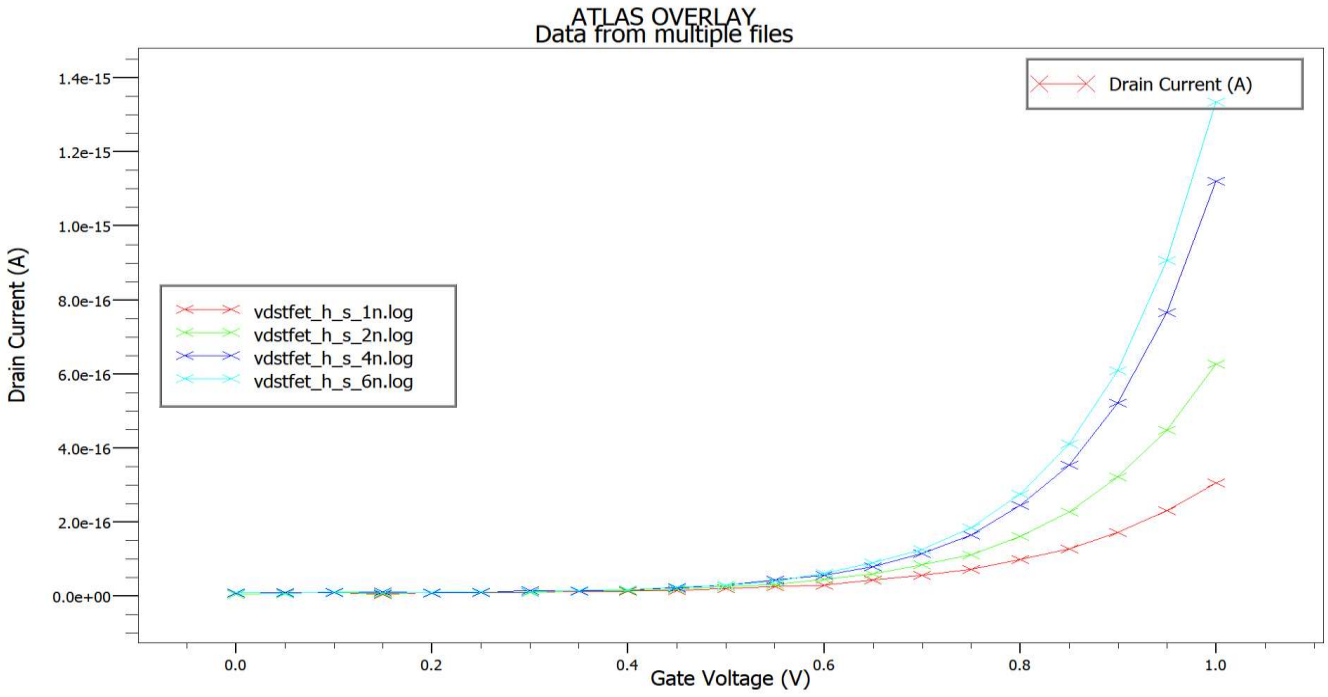
Transfer Characteritics



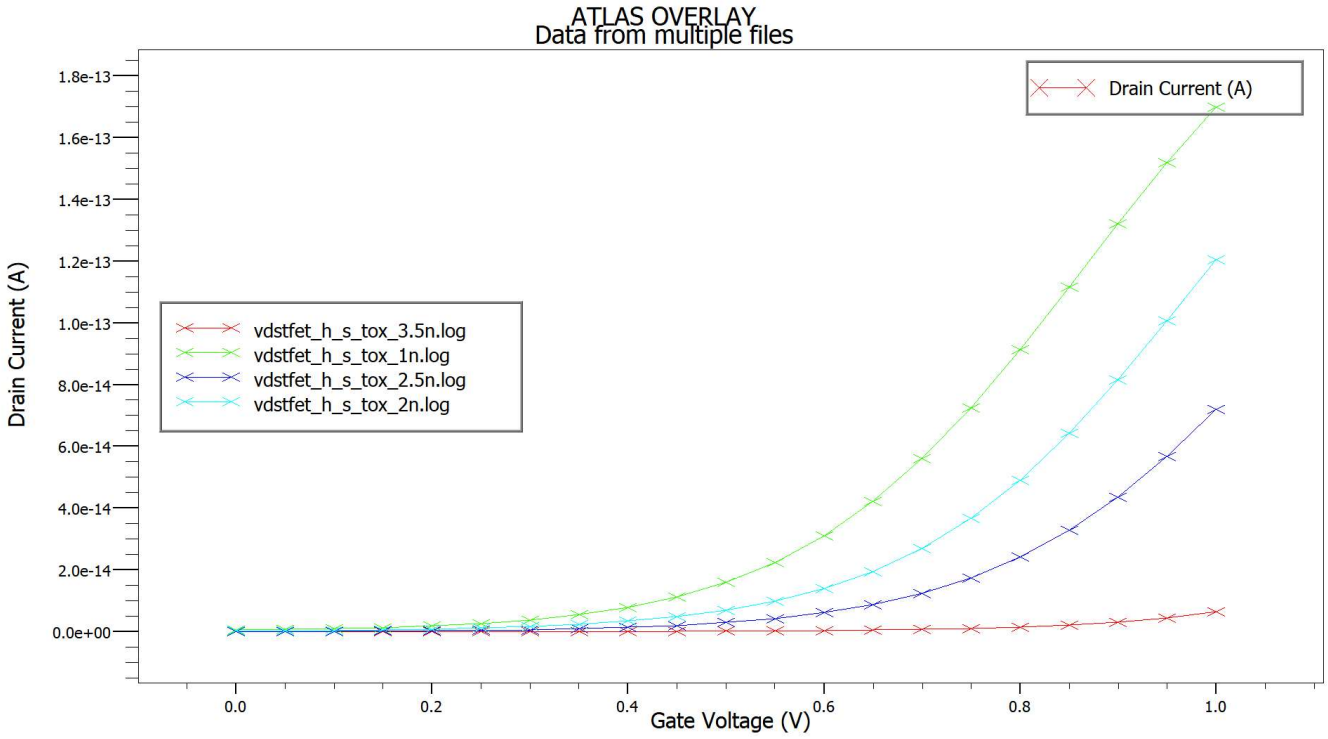
Comparison of Transfer Characteristics for Oxide layer Order



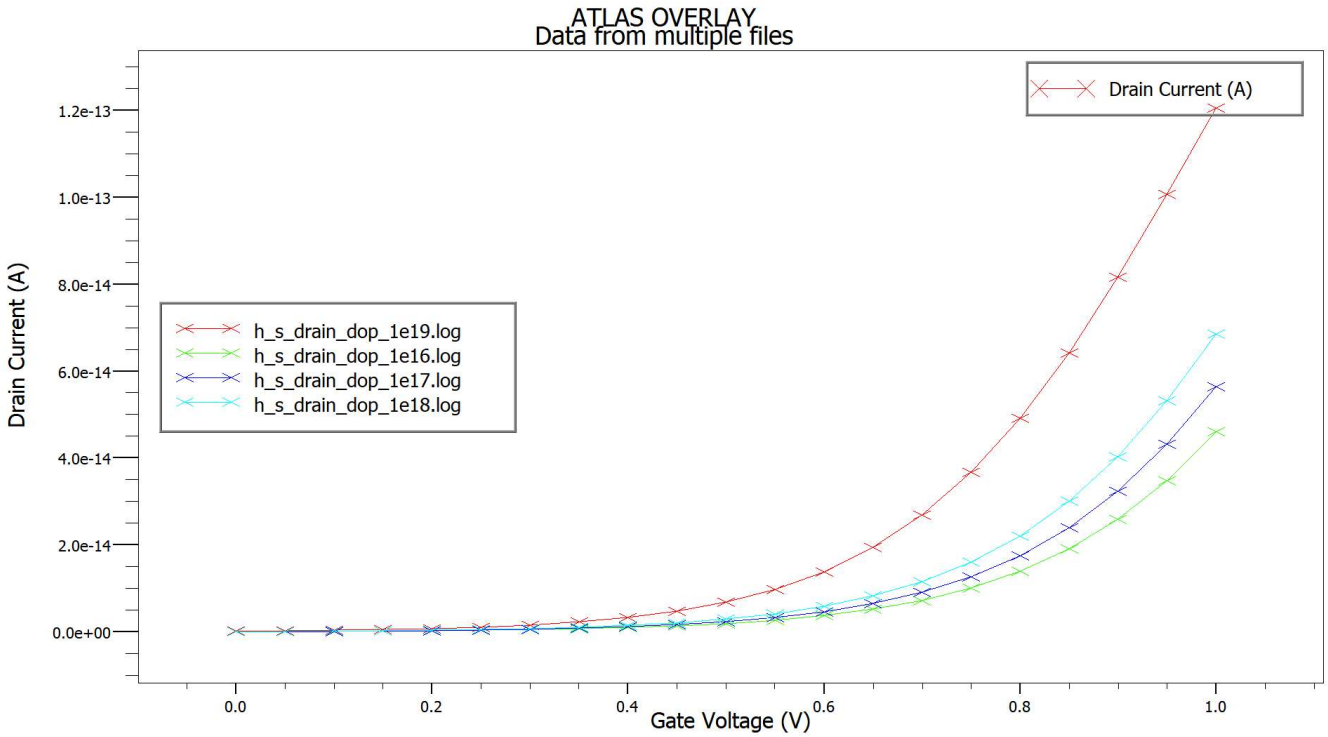
Drain Characteristics



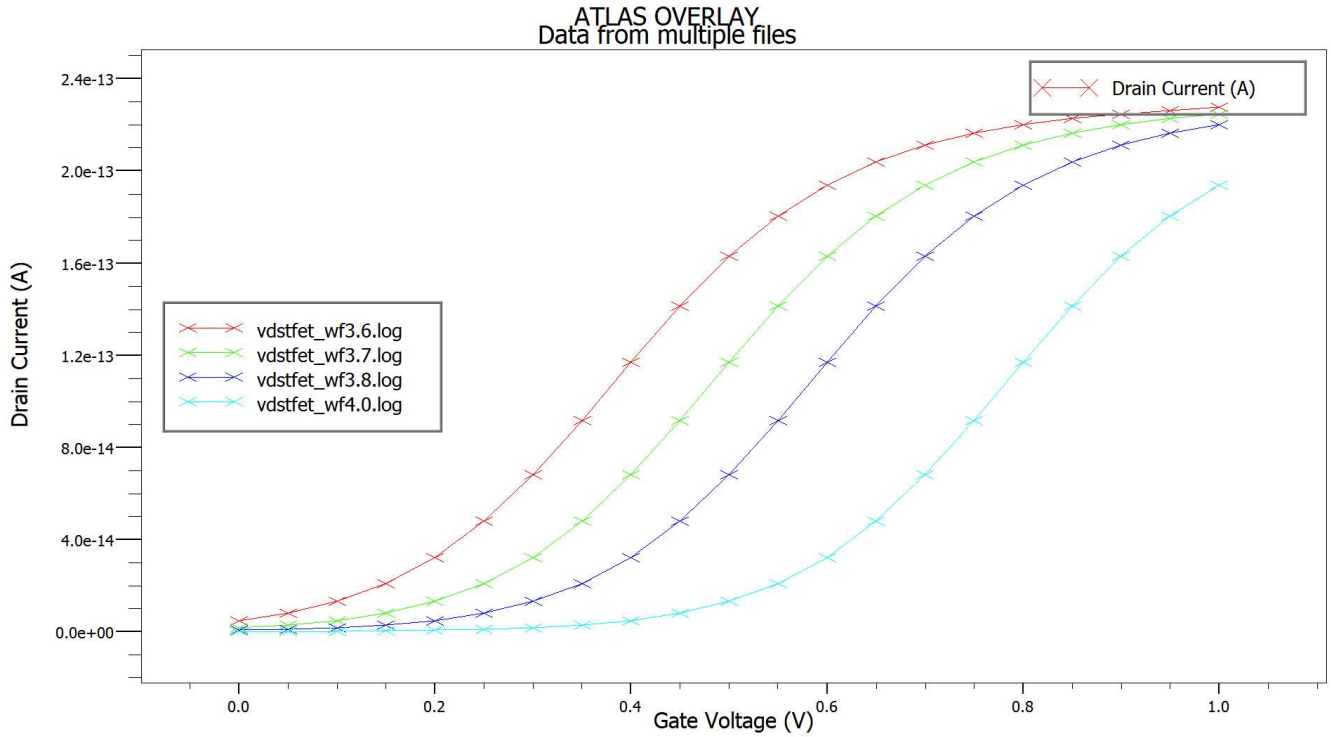
Parameter Variation - Gate Length



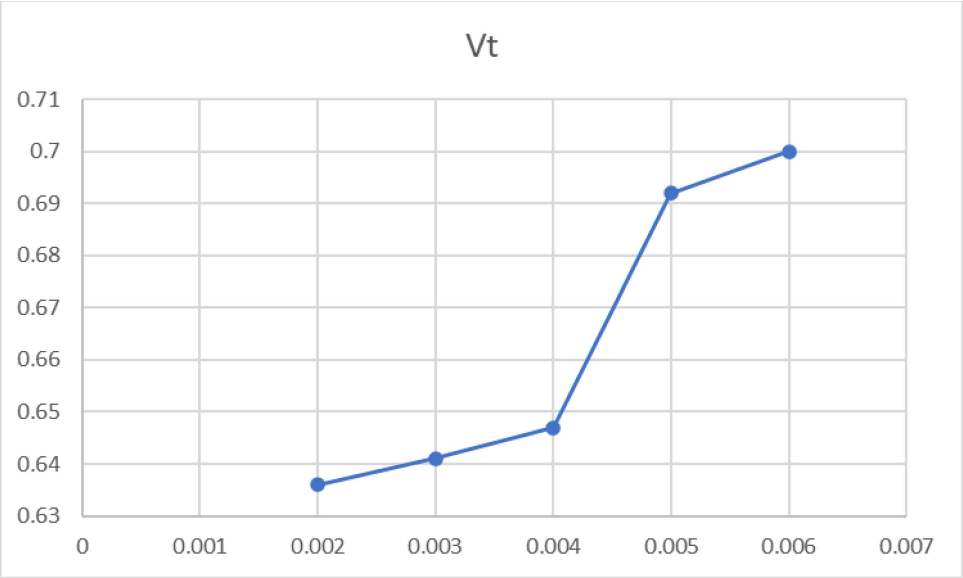
Parameter Variation - Oxide Thickness



Parameter Variation – Drain Doping



Parameter Variation – Gate Work function



Variation in Vt with increase in gate Length

CUT LINE:

DOPING CONCENTRATION:

ADVANTAGES:

1. Provides two tunneling paths, which increases ION current compared to single-source TFETs.
2. Vertical structure enhances gate control over the channel, improving electrostatic behavior.
3. Achieves steep subthreshold swing (<60 mV/dec) enabling ultra-low power switching.
4. Exhibits very low OFF-state leakage current (IOFF), reducing standby power consumption.
5. Reduced Short Channel Effects (SCE) because the channel is vertically aligned and tightly gated.
6. Less Vulnerable to DIBL (Drain Induced Barrier Lowering) due to improved electric field confinement.
7. Capable of reliable operation at low supply voltages (sub-1V), suitable for low-power digital systems.
8. Higher device packing density thanks to vertical stacking — beneficial for future scaling and integration.
9. Better energy efficiency (ION/IOFF ratio) than conventional MOSFETs and many 2D TFETs.
10. Suitable for IoT, wearable, biomedical and battery-operated applications where power efficiency matters

CONCLUSION:

The design and simulation of the Dual Source Vertical TFET demonstrate that introducing an additional source region effectively enhances the band-to-band tunneling rate, resulting in a noticeable improvement in ON-state current while maintaining very low OFF-state leakage. The vertical gate-all-around structure provides strong electrostatic control over the channel, which helps suppress short channel effects and achieve a steep subthreshold slope below 60 mV/dec. Through parameter variation, it was observed that reducing gate length and oxide thickness, optimizing drain doping, and selecting an appropriate gate work function all play a critical role in improving the switching behavior and threshold voltage characteristics. Additionally, the stacked oxide configuration with HfO_2 near the channel and SiO_2 on the outer side offered superior gate coupling and delivered the best overall performance.

FUTURE SCOPE:

The Dual Source Vertical TFET can be further improved through advanced material engineering and structural optimization. Future work can focus on exploring heterojunction source regions, 2D channel materials such as MoS_2 or graphene, and ferroelectric or negative capacitance gate stacks to further enhance tunneling efficiency and reduce threshold voltage. More accurate quantum transport models and temperature-dependent simulations can be incorporated to evaluate real-world operating conditions. Additionally, layout-level optimization, circuit-level integration, and fabrication feasibility studies are needed to transition this device from simulation to practical VLSI systems. With continued refinement, the DSV-TFET holds strong potential for use in next-generation low-power processors, IoT hardware, biomedical implants, and energy-constrained edge computing platforms.